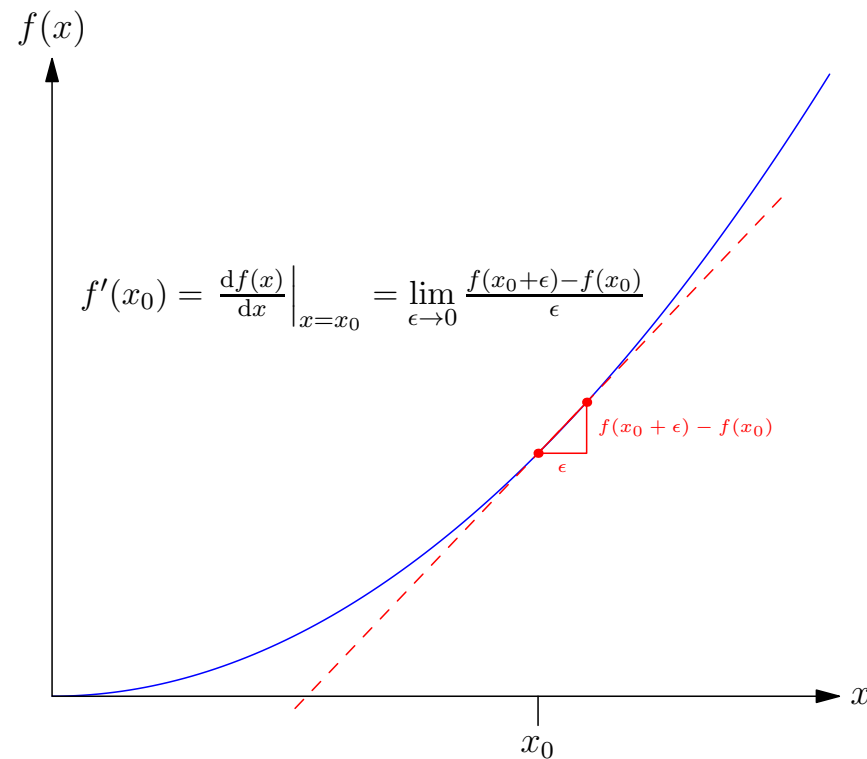


Mathematics of Machine Learning

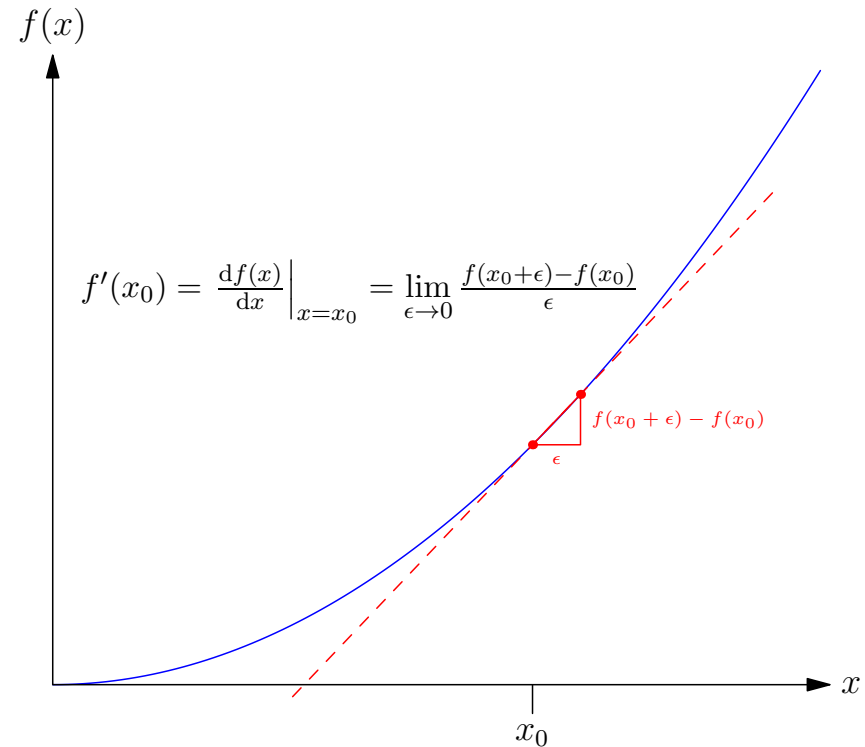
Differential Calculus



Differentiation, product and chain rules, vectors and matrices

Outline

1. **1-D Taylor Expansion?**
2. n -D Taylor Expansion?



Approximating Functions

- In much of machine learning we are trying to approximate functions
- There are many different ways to do this
- Depending on your application some are much better than others
- All function approximation makes some assumptions about the regularity of the function they are approximating
- Taylor expansions are polynomial approximations of a smooth function around a point

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Polynomial Approximation

- Suppose we wanted to approximate a function $f(x)$ around a point x_0 using a polynomial

$$f(x) \approx a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots + a_n(x - x_0)^n$$

- Taking derivatives

$$f'(x) \approx a_1 + 2a_2(x - x_0) + 3a_3(x - x_0)^2 + \dots + na_n(x - x_0)^{n-1}$$

$$f''(x) \approx 2a_2 + 6a_3(x - x_0) + \dots + n(n-1)a_n(x - x_0)^{n-2}$$

- One way to decide on the coefficients is to set $x = x_0$ and assume equality

$$a_0 = f(x_0) \quad a_1 = f'(x_0) \quad a_2 = \frac{1}{2}f''(x_0) \quad a_i = \frac{1}{i!}f^{(i)}(x_0)$$